

Intelligent skiing balance training game system

1. System overview

This is an intelligent skiing balance training system that combines 3D game technology, physics engine, and AI large-scale model analysis. The system provides users with an immersive balance training experience through gamification , and uses AI technology for real-time ability evaluation and personalized training recommendations.

2. Core features

2.1 Competency assessment module

- Comprehensive user information collection (age, gender, balance ability, etc.)
- Multidimensional physical function assessment (single-leg standing, gait stability, etc.)
- Self-Adaptation of difficulty based on evaluation results

2.2 3D game module

- 3D rendering system based on Three.js
- Real physics collision system (Canon.js physics engine)
- Smooth character control and animation system
- Dynamic obstacles and reward systems

2.3 AI analysis module

- Real-time game data collection and analysis
- Personalized training suggestion generation

- Rehabilitation program development based on large models
- Dynamic difficulty adjustment system

3. Game operation guide

3.1 Basic control

- **Left and right movement** : Use the left and right arrow keys or A/D keys on the keyboard to control the skier's turn
- **Balance Control** : Keep the character balanced by moving left and right to avoid falling
- **Speed control** : The system automatically adjusts the initial speed according to the user's ability

3.2 Game Objectives

- **Collect Coins** : Earn points and improve your game performance rating
- **Avoid obstacles** : Avoid obstacles such as trees, rocks, etc
- **Maintain balance** : maintain a stable sliding posture
- **Complete the challenge** : reach the specified distance or collect the target score

3.3 Interface description

- **Score display** : The upper right corner displays the current score in real time
- **Speed Indicator** : Displays the current coasting speed
- **Balance indicator** : shows the current balance status
- **Mileage Count** : Shows distance taxied

3.4 Special operations

- **Sharp turns** : quickly press the arrow keys to achieve emergency avoidance
- **Deceleration** : The system automatically adjusts the speed according to the distribution of obstacles ahead
- **Balance Restoration** : Restore balance quickly through reverse operation

4. Game element features generated by AGI (planned)

4.1 Intelligent Terrain Generation

- **Dynamic Terrain** : Generate random but reasonable ski terrain using AI algorithms
- **Difficulty adaptation** : dynamically adjust terrain complexity based on player performance
- **Scene changes** : intelligently generate different weather effects and light changes
- **Environmental details** : AI automatically adds the right amount of environmental decorative elements

4.2 Intelligent Obstacle System

- **Self-Adaptation distribution** : intelligently adjust obstacle density according to player ability
- **Dynamic difficulty** : Adjust the size and position of obstacles in real time
- **Combination changes** : intelligently generate different combinations of obstacles
- **Predictive Placement** : Predicting Reasonable Obstacle Positions Based on Player Behavior

4.3 Intelligent Reward System

- **Personalized coin path** : Generate the most suitable coin distribution according to player habits
- **Dynamic reward value** : automatically adjust the reward score according to the collection difficulty
- **Achievement Motivation** : AI analyzes player characteristics and sets personalized achievement goals
- **Smart Tips** : Provide timely game tips based on player performance

5. Technical advantages

5.1 Intelligent analysis

- Deep Data Analysis with Large Models

- Provide personalized training advice
- Adjust training difficulty in real time
- Intelligent progress tracking

5.2 Gamification Design

- Immersive 3D gaming experience
- Interesting task design
- Instant feedback system
- Progressive difficulty increase

5.3 Professionalism of balancing assessment and training

- Scientific evaluation system
- Professional training programs
- Safe difficulty control
- Comprehensive data monitoring

6. Advantages of combining with large models

6.1 Intelligent analysis capabilities

- Deep analysis of user data
- Generate personalized suggestions
- Predictive training effect
- Dynamic adjustment scheme

6.2 Natural Language Interaction

- Intelligent feedback system
- Personalized suggestion generation
- Easy-to-understand training instructions

- Humanized interaction experience

6.3 Continuous learning and optimization

- Data accumulation and analysis
- Model continuous optimization
- Training programme update
- Increase in personalization

7. Application value

7.1 Balance Training Medicine

- Provide professional balance training
- Reduce pressure on medical resources
- Improve training effectiveness
- Enhance user compliance

7.2 Health management

- Preventive training
- Health status monitoring
- Long-term health management
- Data tracking analysis

7.3 Sports training

- Professional skills upgrading
- Physical fitness assessment
- Training effect tracking
- Personalized guidance

8. Innovative features

1. **Intelligent evaluation system** : combining large-scale model analysis to provide accurate capability evaluation
2. **Self-Adaptation difficulty adjustment** : dynamically adjust training intensity based on real-time performance
3. **Personalized training program** : Develop exclusive training plan according to user characteristics
4. **Data lake visualization analysis** : visual display of training effect and progress
5. **Gamification Balancing Assessment Experience** : Improving Customer Engagement and Training Motivation

9. Future Outlook

9.1 Technology Upgrade

- **Deep learning integration** : introducing more AI algorithms to improve analysis accuracy
- **MultiModal Machine Learning Interaction** : Add interaction patterns such as speech and gestures
- **Personalized generation of game elements**
- **Device Connection** : Data Acquisition to support more balancing devices
- **Cross-platform support** : extended to mobile end and VR platforms

9.2 Function expansion

- **Social features** : Add multiplayer interaction and competition mechanism
- **Rich Scenes** : Add more training scenes and modes
- **Assessment System** : Improving Competency Assessment Dimensions
- **Smart Tutorial** : Developing Smarter Boot Systems
- **Proprietary large models**: proprietary large models for rehabilitation training

9.3 Application Areas

- **Professional Sports** : Expanding into Professional Sports Training
- **Rehabilitation Centers** : Expanding to More Rehabilitation Institutions
- **Home use** : Develop a version suitable for home use
- **Remote Balancing Training** : Development of Remote Balancing Training System